ST7-TD 5 Multi-Period Model: Step-by-Step Solutions

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1 Exercise 1: Construction of Arbitrage

1.1 Problem Statement

We are given a market with $d = 1$ risky asset and 1 riskless asset. The interest rate is $r = 0$, meaning the riskless asset's price is $S_t^0 = 1$ for all t. We have $T = 2$ periods.

We are given two derivative contracts:

- Contract 1 pays $(S_1)^2$ at time $t = 1$. Its price today ($t = 0$) is p_1.
- Contract 2 pays $(S_2)^2$ at time $t = 2$. Its price today ($t = 0$) is p_2.

We assume $p_1 > p_2$. Our goal is to construct a self-financing portfolio with an initial cost of zero ($V_0 = 0$) that guarantees a strictly positive profit at the end ($V_2 > 0$).

1.2 Step-by-Step Strategy

Time $t = 0$: Initializing the Portfolio

Since $p_1 > p_2$, the claim paying out at $t = 1$ is more expensive than the claim paying out at $t = 2$. The core idea of arbitrage is to "sell high and buy low".

1. **Sell (Short) Contract 1:** We sell the payoff $(S_1)^2$. This generates a positive cash flow of $+p_1$ today.
2. **Buy Contract 2:** We buy the payoff $(S_2)^2$. This costs us $-p_2$ today.
3. **Bank Account:** We have a net positive cash flow of $p_1 - p_2 > 0$. We invest this into the riskless asset S^0. Because $r = 0$, this cash simply sits in the account without generating interest.

The initial cost of this portfolio is exactly 0, so $V_0 = 0$.

Time $t = 1$: Rebalancing

At $t = 1$, Contract 1 expires and we are obligated to pay out $(S_1)^2$. Let's look at our cash (riskless asset) position right after paying this obligation:

$$\text{Cash} = (p_1 - p_2) - (S_1)^2 \tag{1}$$

At this point, we need to make a strategic trade to guarantee a profit at $t = 2$. We know that at $t = 2$, we will receive $(S_2)^2$ from Contract 2. To ensure our final wealth is positive, we want to construct a perfect square: $(S_1 - S_2)^2 = (S_1)^2 - 2S_1S_2 + (S_2)^2$.

Right now, we are missing the $+2(S_1)^2$ and the $-2S_1S_2$ terms. To create them, we do the following:

1. **Short the Risky Asset:** We short a quantity of $2S_1$ shares of the risky asset. Since the price of one share at $t = 1$ is S_1, shorting $2S_1$ shares generates a cash inflow of $2S_1 \times S_1 = +2(S_1)^2$.
2. **Update Bank Account:** We deposit this new cash into our riskless asset account. Our new cash balance becomes:
$$\text{Cash} = [(p_1 - p_2) - (S_1)^2] + 2(S_1)^2 = (p_1 - p_2) + (S_1)^2 \tag{2}$$

Our portfolio heading into $t = 2$ now consists of:

- $(p_1 - p_2) + (S_1)^2$ invested in the riskless asset.
- A short position of $2S_1$ shares of the risky asset.
- A long position in Contract 2, which will pay $(S_2)^2$.

Time $t = 2$: Final Payoff

At $t = 2$, we liquidate everything:

1. We receive the payoff from Contract 2: $+(S_2)^2$.
2. We must close our short position in the risky asset. We owe $2S_1$ shares, and the price is now S_2. Buying them back costs us $-2S_1S_2$.
3. We withdraw our cash from the riskless asset: $+(p_1 - p_2) + (S_1)^2$.

Let's sum these up to find the final value of our portfolio, V_2:

$$V_2 = (S_2)^2 - 2S_1S_2 + (p_1 - p_2) + (S_1)^2 \tag{3}$$

$$V_2 = \left((S_1)^2 - 2S_1S_2 + (S_2)^2\right) + (p_1 - p_2) \tag{4}$$

$$V_2 = (S_1 - S_2)^2 + (p_1 - p_2) \tag{5}$$

1.3 Conclusion

We have $(S_1 - S_2)^2 \geq 0$ for any possible price path. Furthermore, we assumed $p_1 - p_2 > 0$. Therefore, $V_2 \geq p_1 - p_2 > 0$.

Since our strategy cost nothing to initiate ($V_0 = 0$) and guarantees a strictly positive payoff without any risk of loss, it is an arbitrage opportunity.

2 Exercise 2: Linear Payoff

2.1 Problem Statement

We want to replicate a specific payoff H, and we do *not* assume the market is arbitrage-free. First, we need to show that a payoff depending on a single asset at a specific past or present date, $H = S^i_{t^*}$, is replicable at maturity T. Then, we will generalize this to show that any linear combination of asset prices across different times is also replicable.

2.2 Part 1: Replicating H = S_t i

The payoff H is paid at the final time T, but its value is completely determined and known at time t^* (it is $\mathcal{F}_{t^*}$-measurable).

To replicate this, our strategy will have two phases:

1. **Phase 1 (From $t = 1$ to t^*):** We hold a constant fraction of the risky asset i so that at time t^*, its value is exactly the present value of the payoff we need at time T.

2. **Phase 2 (From $t^* + 1$ to T):** At time t^*, we sell the risky asset and put all the proceeds into the riskless bank account S^0, letting it accrue interest until time T.

Defining the Strategy mathematically

Let's build the portfolio $\phi_t = (\xi^0_t, \xi^1_t, \ldots, \xi^d_t)$ for $t \in \{1, \ldots, T\}$.

For $1 \leq t \leq t^*$ (Phase 1):
We invest solely in the risky asset i. We want to end up with $S^i_{t^*}$ at time T. Discounting this back to time t^*, we need a value of $\frac{S^i_{t^*}}{(1+r)^{T-t^*}}$ at time t^*. Since the price of asset i at time t^* is $S^i_{t^*}$, we need exactly $\frac{1}{(1+r)^{T-t^*}}$ shares.

- $\xi^i_t = \frac{1}{(1+r)^{T-t^*}}$
- $\xi^0_t = 0$ (no money in the bank)
- $\xi^j_t = 0$ for all $j \neq i, 0$

For $t^* + 1 \leq t \leq T$ (Phase 2):
At time t^*, we liquidate the risky asset. The value of our portfolio at time t^* is:

$$V_{t^*} = \xi^i_{t^*} S^i_{t^*} = \frac{S^i_{t^*}}{(1+r)^{T-t^*}} \tag{6}$$

We use all this value to buy units of the riskless asset $\xi^0_{t^*+1}$. The price of the riskless asset at time t^* is $S^0_{t^*} = (1+r)^{t^*}$. Therefore, the number of units we buy is:

$$\xi^0_{t^*+1} = \frac{V_{t^*}}{S^0_{t^*}} = \frac{\frac{S^i_{t^*}}{(1+r)^{T-t^*}}}{(1+r)^{t^*}} = \frac{S^i_{t^*}}{(1+r)^T} \tag{7}$$

So, for the remainder of the time until T, our positions are:

- $\xi^0_t = \frac{S^i_{t^*}}{(1+r)^T}$
- $\xi^j_t = 0$ for all $j \neq 0$ (no risky assets)

Checking the Self-Financing Condition

A strategy is self-financing if changes in the portfolio composition do not require injecting or withdrawing external cash. This must hold at our rebalancing time t^*:

$$(\xi^0_{t^*+1} - \xi^0_{t^*})S^0_{t^*} + \sum_{j=1}^{d}(\xi^j_{t^*+1} - \xi^j_{t^*})S^j_{t^*} = 0 \tag{8}$$

Substituting our chosen quantities:

$$\left(\frac{S^i_{t^*}}{(1+r)^T} - 0\right)(1+r)^{t^*} + \left(0 - \frac{1}{(1+r)^{T-t^*}}\right)S^i_{t^*} = 0 \tag{9}$$

$$\frac{S^i_{t^*}}{(1+r)^{T-t^*}} - \frac{S^i_{t^*}}{(1+r)^{T-t^*}} = 0 \tag{10}$$

The condition holds. The strategy is predictable, self-financing, and at time T, the value is:

$$V_T = \xi^0_T S^0_T = \left(\frac{S^i_{t^*}}{(1+r)^T}\right)(1+r)^T = S^i_{t^*} = H \tag{11}$$

Thus, $H = S^i_{t^*}$ is replicable.

2.3 Part 2: Replicating a General Linear Payoff

We now consider a general linear payoff of the form:

$$H = \sum_{k=0}^{T}\sum_{i=1}^{d}\alpha(i,k)S^i_k \tag{12}$$

where $\alpha(i,k)$ are constant coefficients.

Let $\phi(i,k)_t = (\xi^0_t(i,k), \xi_t(i,k))$ be the self-financing strategy that replicates the single payoff S^i_k, which we just proved exists in Part 1.

Because the financial market operations (portfolio value and self-financing condition) are strictly linear, any linear combination of payoffs can be replicated by the exact same linear combination of their respective replicating strategies.

Therefore, the payoff H is replicable by the composite strategy Φ_t:

$$\Phi_t = \sum_{k=0}^{T}\sum_{i=1}^{d}\alpha(i,k)\phi(i,k)_t \tag{13}$$

This strategy simply means holding $\alpha(i,k)$ units of the portfolio $\phi(i,k)_t$ for every i and k. Since each sub-portfolio is predictable and self-financing, their linear combination remains predictable and self-financing, successfully replicating H.

3 Exercise 3: Put-Call Parity

3.1 Problem Statement

We assume a market with one risky asset ($d = 1$) that is free of arbitrage opportunities. Let C_0 and P_0 be the prices at time $t = 0$ of a European call and a European put, respectively, both with the same strike price K and the same maturity date T.

We want to prove the fundamental put-call parity relationship:

$$C_0 - P_0 = S_0 - \frac{K}{(1+r)^T} \tag{14}$$

3.2 Step-by-Step Proof using No-Arbitrage

To prove this relationship, we will construct two different investment strategies at time $t = 0$. We will show that both strategies yield the exact same payoff at time T, regardless of what the final stock price S_T is. By the Law of One Price (a consequence of no-arbitrage), two portfolios with identical future payoffs must have the exact same initial cost.

Strategy 1: Options Portfolio

- **Action at $t = 0$:** We short (sell) one call option and long (buy) one put option.
- **Initial Value $V_0^{(1)}$:** Selling the call brings in C_0, and buying the put costs P_0. The net value of this portfolio today is $-C_0 + P_0$.
- **Payoff at $t = T$:**
  - The short call pays $-(S_T - K)_+$
  - The long put pays $+(K - S_T)_+$

  Let's evaluate the combined payoff $V_T^{(1)}$ for any possible value of S_T:

$$V_T^{(1)} = -(S_T - K)_+ + (K - S_T)_+ \tag{15}$$

$$= \begin{cases} -(S_T - K) + 0 = K - S_T & \text{if } S_T \geq K \\ 0 + (K - S_T) = K - S_T & \text{if } S_T < K \end{cases} \tag{16}$$

  In all cases, $V_T^{(1)} = K - S_T$.

Strategy 2: Underlying and Riskless Asset

- **Action at $t = 0$:** We short one share of the underlying asset and invest the discounted strike price, $\frac{K}{(1+r)^T}$, into the riskless bank account.
- **Initial Value $V_0^{(2)}$:** Shorting the share gives us a negative position value of $-S_0$. The money in the bank has a value of $+\frac{K}{(1+r)^T}$. The net value of this portfolio today is $-S_0 + \frac{K}{(1+r)^T}$.
- **Payoff at $t = T$:**
  - We must buy back the shorted share, which will cost us $-S_T$.
  - The money in the bank account has grown with interest: $\frac{K}{(1+r)^T} \times (1+r)^T = K$.

  The combined payoff is $V_T^{(2)} = K - S_T$.

3.3 Conclusion

We have established that both strategies yield the exact same final payoff:

$$V_T^{(1)} = V_T^{(2)} = K - S_T \tag{17}$$

Because the market is arbitrage-free, two portfolios that always have the same final value must have the exact same initial value. Therefore, $V_0^{(1)} = V_0^{(2)}$:

$$-C_0 + P_0 = -S_0 + \frac{K}{(1+r)^T} \tag{18}$$

Multiplying the entire equation by -1 gives us the desired put-call parity formula:

$$C_0 - P_0 = S_0 - \frac{K}{(1+r)^T} \tag{19}$$

4 Exercise 4: Dependency of the Call Price on the Strike

4.1 Problem Statement

We assume a market with one risky asset ($d = 1$) that is arbitrage-free and complete. We want to study the properties of the arbitrage-free price of a European call option, denoted as $C_t(K, T)$, with respect to its strike price K and its maturity T.

4.2 Part 1: Monotonicity of Prices based on Payoffs

Question: If H_1 and H_2 are two payoffs satisfying $H_1 \leq H_2$ $\mathbb{P}$-almost surely, show that their arbitrage-free prices at any date satisfy the same inequality.

Step-by-Step Proof:

1. Because the market is complete and arbitrage-free, the Fundamental Theorems of Asset Pricing state that there exists a unique equivalent martingale measure (risk-neutral measure) $\mathbb{Q}$.

2. Under this measure $\mathbb{Q}$, the arbitrage-free price of any replicable payoff H at time t is given by the conditional expectation of its discounted final payoff:

$$V_t(H) = \mathbb{E}_{\mathbb{Q}}\left[\frac{H}{(1+r)^{T-t}}\middle|\mathcal{F}_t\right] \tag{20}$$

3. We are given that $H_1 \leq H_2$ $\mathbb{P}$-a.s. Since $\mathbb{P}$ and $\mathbb{Q}$ are equivalent measures, they agree on events of probability zero, meaning $H_1 \leq H_2$ $\mathbb{Q}$-a.s. as well.

4. The conditional expectation is a positive linear operator. Therefore, it preserves inequalities. Since $H_1 \leq H_2$ and $(1+r)^{T-t} > 0$, it immediately follows that:

$$\mathbb{E}_{\mathbb{Q}}\left[\frac{H_1}{(1+r)^{T-t}}\middle|\mathcal{F}_t\right] \leq \mathbb{E}_{\mathbb{Q}}\left[\frac{H_2}{(1+r)^{T-t}}\middle|\mathcal{F}_t\right] \tag{21}$$

5. Substituting our pricing formula, we get $V_t(H_1) \leq V_t(H_2)$.

4.3 Part 2: The Call Price is Non-Increasing with Respect to Strike

Question: Deduce that $K \mapsto C_t(K, T)$ is non-increasing.

Step-by-Step Proof:

1. Let K_1 and K_2 be two strike prices such that $K_1 \leq K_2$. We want to show that $C_t(K_1, T) \geq C_t(K_2, T)$.

2. The payoff of a call option with strike K is $H(K) = (S_T - K)_+ = \max(S_T - K, 0)$.

3. Since $K_1 \leq K_2$, we have $-K_1 \geq -K_2$, and thus $S_T - K_1 \geq S_T - K_2$.

4. Applying the maximum function preserves this inequality: $\max(S_T - K_1, 0) \geq \max(S_T - K_2, 0)$.

5. Therefore, the payoffs satisfy $H(K_1) \geq H(K_2)$ for all possible states of the world.

6. Using the result from Part 1, since the payoff of call 1 is always greater than or equal to call 2, its price must also be greater than or equal: $C_t(K_1, T) \geq C_t(K_2, T)$. The function is non-increasing.

4.4 Part 3: Proving Monotonicity via No-Arbitrage Strategies

Question: Using the definition of no-arbitrage, prove the result above (this approach does not require the market to be complete).

Step-by-Step Proof:

1. Assume $K_1 \leq K_2$. Consider a portfolio constructed at time t by buying one call with strike K_1 and selling (shorting) one call with strike K_2.
2. The value of this portfolio at maturity T is $V_T = (S_T - K_1)_+ - (S_T - K_2)_+$.
3. Let's analyze this payoff V_T for all possible values of S_T:
   - If $S_T \leq K_1 \leq K_2$: Both options expire worthless. $V_T = 0 - 0 = 0$.
   - If $K_1 < S_T \leq K_2$: The first option pays out, the second does not. $V_T = (S_T - K_1) - 0 > 0$.
   - If $K_1 \leq K_2 < S_T$: Both options pay out. $V_T = (S_T - K_1) - (S_T - K_2) = K_2 - K_1 \geq 0$.
4. In all scenarios, $V_T \geq 0$.
5. By the principle of no-arbitrage, if a portfolio has a guaranteed non-negative payoff at maturity, its cost at any prior time t must also be non-negative. If it cost less than 0, buying it would be an arbitrage opportunity.
6. Therefore, $V_t \geq 0 \implies C_t(K_1, T) - C_t(K_2, T) \geq 0 \implies C_t(K_1, T) \geq C_t(K_2, T)$.

4.5 Part 4: Convexity of the Call Price

Question: Show that $K \mapsto C_t(K, T)$ is convex.

Step-by-Step Proof:

1. Let K_1, K_2 be two strike prices, and let $\lambda \in [0, 1]$. Define a new strike price as a convex combination: $K = \lambda K_1 + (1 - \lambda)K_2$.
2. The function $f(x) = (x)_+ = \max(x, 0)$ is a convex function.
3. We apply the definition of convexity to the option payoff at time T:

$$(S_T - K)_+ = (S_T - (\lambda K_1 + (1 - \lambda)K_2))_+ \tag{22}$$
$$= (\lambda S_T - \lambda K_1 + (1 - \lambda)S_T - (1 - \lambda)K_2)_+ \tag{23}$$
$$= (\lambda(S_T - K_1) + (1 - \lambda)(S_T - K_2))_+ \tag{24}$$

4. By the convexity of the maximum function:

$$(\lambda(S_T - K_1) + (1 - \lambda)(S_T - K_2))_+ \leq \lambda(S_T - K_1)_+ + (1 - \lambda)(S_T - K_2)_+ \tag{25}$$

5. This inequality holds for the payoffs. By taking the conditional expectation under the risk-neutral measure (as justified in Part 1), the inequality is preserved for the prices:

$$C_t(K, T) \leq \lambda C_t(K_1, T) + (1 - \lambda)C_t(K_2, T) \tag{26}$$

6. This is the exact definition of a convex function.

4.6 Part 5: Monotonicity with Respect to Maturity

Question: Assuming $r = 0$, show that $T \mapsto C_0(K, T)$ is non-decreasing.

Step-by-Step Proof:

1. We want to compare the price of a call expiring at T with one expiring at $T + 1$. Since $r = 0$, the discount factor $(1 + r)^{-t}$ is just 1.

2. The price of the option maturing at $T + 1$ is $C_0(K, T + 1) = \mathbb{E}_{\mathbb{Q}}[(S_{T+1} - K)_+]$.

3. By the Law of Iterated Expectations (Tower Property), we can condition on the information available at time T:
$$\mathbb{E}_{\mathbb{Q}}[(S_{T+1} - K)_+] = \mathbb{E}_{\mathbb{Q}}\left[\mathbb{E}_{\mathbb{Q}}[(S_{T+1} - K)_+ \mid \mathcal{F}_T]\right] \tag{27}$$

4. Consider the inner expectation: $\mathbb{E}_{\mathbb{Q}}[(S_{T+1} - K)_+ \mid \mathcal{F}_T]$. The function $g(x) = (x - K)_+$ is convex.

5. Jensen's Inequality states that for a convex function g, $\mathbb{E}[g(X)] \geq g(\mathbb{E}[X])$. Applying this to our inner expectation:
$$\mathbb{E}_{\mathbb{Q}}[(S_{T+1} - K)_+ \mid \mathcal{F}_T] \geq (\mathbb{E}_{\mathbb{Q}}[S_{T+1} \mid \mathcal{F}_T] - K)_+ \tag{28}$$

6. Under the risk-neutral measure $\mathbb{Q}$ and with $r = 0$, the discounted stock price is a martingale. This means $\mathbb{E}_{\mathbb{Q}}[S_{T+1} \mid \mathcal{F}_T] = S_T$. Substituting this back:
$$\mathbb{E}_{\mathbb{Q}}[(S_{T+1} - K)_+ \mid \mathcal{F}_T] \geq (S_T - K)_+ \tag{29}$$

7. Now, we take the unconditional expectation of both sides (from step 3):
$$\mathbb{E}_{\mathbb{Q}}\left[\mathbb{E}_{\mathbb{Q}}[(S_{T+1} - K)_+ \mid \mathcal{F}_T]\right] \geq \mathbb{E}_{\mathbb{Q}}[(S_T - K)_+] \tag{30}$$

8. The left side is exactly $C_0(K, T + 1)$ and the right side is exactly $C_0(K, T)$. Therefore:
$$C_0(K, T + 1) \geq C_0(K, T) \tag{31}$$

9. Thus, the call price is non-decreasing with respect to maturity.

5 Exercise 5: Trinomial Model

5.1 Problem Statement

We consider a discrete-time market model where the return of the risky asset, defined as $R_t := \frac{S_t - S_{t-1}}{S_{t-1}}$, can take one of three possible values at each time step: a, b, or c, such that $-1 < a < b < c$. The returns are independent and identically distributed (i.i.d.) with strictly positive probabilities $p_a, p_b, p_c > 0$. This market consists of one riskless asset (with constant interest rate r) & one risky asset.

5.2 Part 1: Arbitrage-Free Condition (a < r < c)

Question: Show that if the market is arbitrage-free, then $a < r < c$.

Step-by-Step Proof: We will prove this by contradiction. We will show that if $r \geq c$, we can construct an arbitrage opportunity. A symmetric argument applies if $r \leq a$.

Case 1: Assume $r \geq c$

1. **Initial Strategy ($t = 0$):** We short one unit of the risky asset. This gives us $+S_0$ in cash. We take this S_0 and invest it entirely in the riskless bank account.
2. **Initial Cost:** The initial value of this portfolio is exactly 0.
3. **Value at Maturity ($t = T$):**
   - The money in the bank has grown to $S_0(1+r)^T$.
   - We must buy back the shorted asset, which costs $-S_T$.
4. **Analyzing the Payoff:** The final value is $V_T = S_0(1+r)^T - S_T$.
5. We know that the maximum possible return in any single period is c. Therefore, the absolute maximum price the stock can reach at time T is if it returns c every single period: $S_T \leq S_0(1+c)^T$.
6. Substituting this maximum stock price into our payoff equation gives the worst-case scenario for our portfolio:
$$V_T \geq S_0(1+r)^T - S_0(1+c)^T$$
7. Since we assumed $r \geq c$, it follows that $(1+r)^T \geq (1+c)^T$. Therefore:
$$S_0(1+r)^T - S_0(1+c)^T \geq 0 \implies V_T \geq 0$$
   Our portfolio will never lose money.
8. **Strict Positivity:** Because $b < c$ and the probability of outcome b is strictly positive ($p_b > 0$), there is a non-zero probability that the stock price at T will be strictly less than $S_0(1+c)^T$. In these states, $V_T > 0$.
9. **Conclusion for Case 1:** A portfolio costing zero today that never loses money and has a positive probability of making money is an arbitrage opportunity. To avoid this, we must have $r < c$.

Case 2: Assume $r \leq a$

1. **Strategy:** We do the exact opposite. We borrow money from the bank (short the riskless asset) to buy the risky asset.
2. **Analyzing the Payoff:** At $t = T$, the worst possible return for the stock is a. The lowest possible stock price is $S_0(1+a)^T$.
3. The final value is $V_T = S_T - S_0(1+r)^T$.
4. In the worst-case scenario, $V_T \geq S_0(1+a)^T - S_0(1+r)^T$. Since $r \leq a$, this payoff is always ≥ 0.

5. Again, because $p_b > 0$ and $b > a$, there is a positive probability of making a strict profit. This is an arbitrage.

6. **Conclusion for Case 2:** To avoid this arbitrage, we must have $r > a$.

Combining both cases, an arbitrage-free market requires $a < r < c$.

5.3 Part 2: Market Completeness

Question: Is the market complete?
Step-by-Step Proof:

1. A market is complete if and only if every possible payoff can be perfectly replicated by a self-financing portfolio of the available traded assets.

2. Let's look at a single time step, $T = 1$. At time $t = 1$, there are exactly 3 possible states of the world for the stock price:

   - State 1: $S_1 = S_0(1 + a)$
   - State 2: $S_1 = S_0(1 + b)$
   - State 3: $S_1 = S_0(1 + c)$

3. Because there are 3 states, any arbitrary derivative payoff H at $t = 1$ can be represented as a 3-dimensional vector in $\mathbb{R}^3$: $H = (H_a, H_b, H_c)^T$.

4. To replicate this payoff, we hold Δ units of the risky asset and B units of the riskless asset. Our portfolio's payoff vector across the 3 states will be:

$$\Delta \begin{pmatrix} S_0(1+a) \\ S_0(1+b) \\ S_0(1+c) \end{pmatrix} + B \begin{pmatrix} 1+r \\ 1+r \\ 1+r \end{pmatrix}$$

5. This equation describes a linear combination of two fixed vectors. The set of all possible linear combinations of two vectors forms a 2-dimensional plane (a subspace) within the 3-dimensional space $\mathbb{R}^3$.

6. Since a 2-dimensional plane cannot cover the entirety of a 3-dimensional space, there exist payoff vectors $H \in \mathbb{R}^3$ that simply do not lie on this plane.

7. Because there are payoffs that cannot be generated by any combination of Δ and B, these payoffs are not replicable.

8. Therefore, the trinomial market is **incomplete**.